

Comparative Analysis of Churn Prediction Models

A Technical Evaluation Report

Prepared by: Data Science Team

Subject: Telco Customer Churn Strategy

Executive Summary

This report compares three machine learning architectures for predicting customer churn. The business objective is to maximize the identification of at-risk customers (Recall for Churners) to enable proactive retention campaigns. We evaluate a baseline Random Forest classifier, a class-weighted Logistic Regression model, and an advanced Deep Learning architecture (Multi-Layer Perceptron).

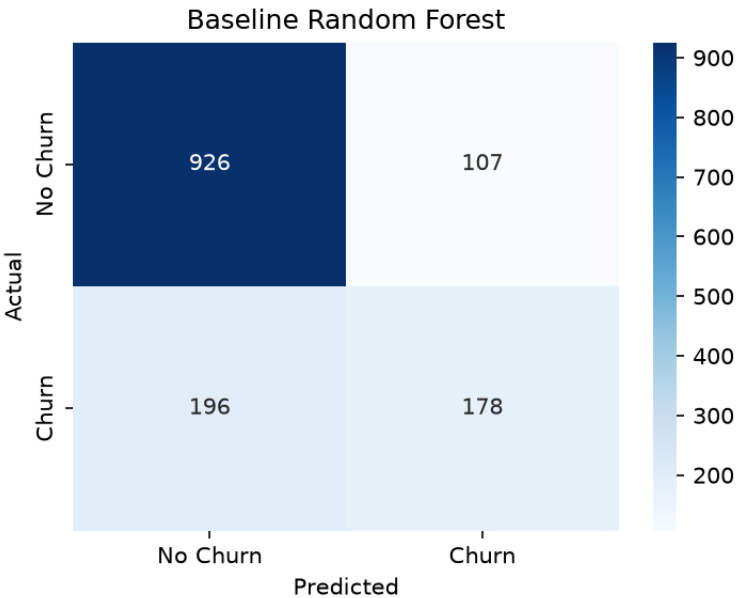
Model 1: Baseline Random Forest

Overall Accuracy: 78.46%

The baseline model achieves high accuracy but suffers from severe class imbalance, predicting 'No Churn' too aggressively.

Churn Recall: Only 47.59% of actual churners were identified.

Class	Precision	Recall	F1-Score	Support
Class 0 (No Churn)	0.83	0.90	0.86	1033.0
Class 1 (Churn)	0.62	0.48	0.54	374.0
Macro Avg	0.72	0.69	0.70	1407.0
Weighted Avg	0.77	0.78	0.77	1407.0



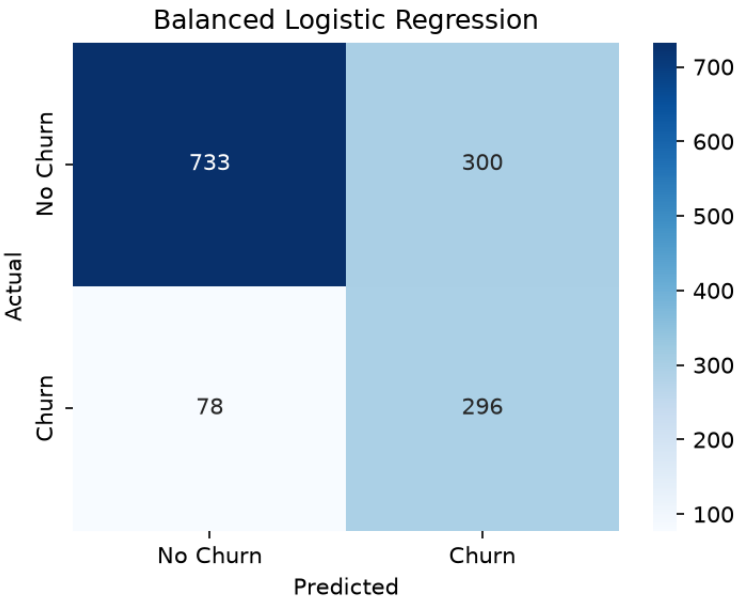
Model 2: Balanced Logistic Regression

Overall Accuracy: 73.13%

By applying 'balanced' class weights, this linear model heavily penalizes the misclassification of the minority class (Churners).

Churn Recall: 79.14% of actual churners were successfully identified.

Class	Precision	Recall	F1-Score	Support
Class 0 (No Churn)	0.90	0.71	0.80	1033.0
Class 1 (Churn)	0.50	0.79	0.61	374.0
Macro Avg	0.70	0.75	0.70	1407.0
Weighted Avg	0.80	0.73	0.75	1407.0



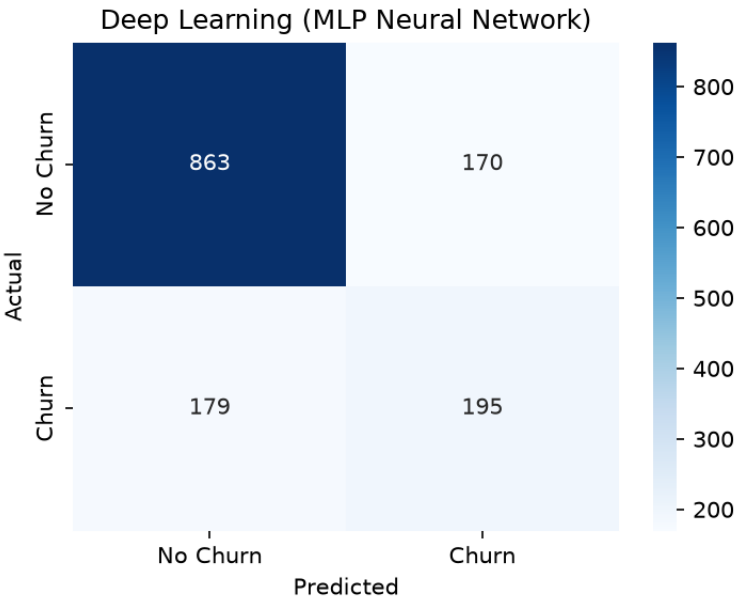
Model 3: Advanced Deep Learning (MLP)

Overall Accuracy: 75.20%

This advanced deep learning architecture utilizes multiple hidden layers to capture highly complex, non-linear feature interactions in the customer data.

Churn Recall: 52.14% of actual churners were successfully identified.

Class	Precision	Recall	F1-Score	Support
Class 0 (No Churn)	0.83	0.84	0.83	1033.0
Class 1 (Churn)	0.53	0.52	0.53	374.0
Macro Avg	0.68	0.68	0.68	1407.0
Weighted Avg	0.75	0.75	0.75	1407.0



Strategic Recommendation

The baseline Random Forest misses over half of the churning customers. While the Logistic Regression model captures 79.14% of churners, the Deep Learning model captures 52.14% of churners while maintaining a robust F1-score of 0.53. Depending on the specific cost of false positives vs. false negatives, either Logistic Regression or the Deep Learning model provides a massive upgrade over the baseline.